

# Cheat Sheet Materi 5

TI5001 / Optimisasi & Intelegensia Komputasional · Pemecahan Analitik Model Optimisasi Tak-Linier Himpunan Berpembatas (Persamaan dan Pertidaksamaan)

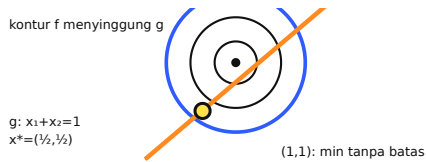
## 1 · Pembatas persamaan: Lagrange

Opt  $f(x)$ ,  $x \in \mathbb{R}^n$ , dengan  $m$  pembatas  $g_i(x)=0$  (f dapat diturunkan 2x,  $g_i$  dapat diturunkan)

$$L(x, \lambda) = f(x) - \sum_{i=1..m} \lambda_i g_i(x)$$

$\lambda$  = pengali Lagrange (m buah)

**Kondisi perlu:**  $\partial L / \partial x_j = 0$  ( $j=1..n$ ) dan  $\partial L / \partial \lambda_i = -g_i(x) = 0$  ( $i=1..m$ )  $\rightarrow n+m$  persamaan untuk  $n+m$  variabel ( $x, \lambda$ )



Jika ditulis  $L=f+\lambda g: x^*$  tetap, hanya tanda  $\lambda$  berbalik.

## 2 · Contoh Lagrange

**Slide:** min  $x_1^2 + x_2^2 - 2x_1 - 2x_2$ ,  $x_1 + x_2 = 1$   
 $2x_1 - 2 - \lambda = 0$ ;  $2x_2 - 2 - \lambda = 0$ ;  $x_1 + x_2 = 1 \rightarrow x^* = (1/2, 1/2)$ ,  $\lambda = -1$ ,  $f = -3/2$   
 maks  $x_1 x_2$ ,  $x_1 + x_2 = 10$ :  $x_2 - \lambda = 0$ ,  $x_1 - \lambda = 0 \rightarrow x_1 = x_2 = \lambda = 5$ ,  $f = 25$   
 min  $x_1^2 + x_2^2$ ,  $2x_1 + x_2 = 5$ :  $x_1 = \lambda$ ,  $x_2 = \lambda/2 \rightarrow \lambda = 2$ ,  $x^* = (2, 1)$ ,  $f = 5$   
 maks  $x_1 + x_2$ ,  $x_1^2 + x_2^2 = 2$ :  $x_1 = x_2 = 1/(2\lambda)$ ,  $\lambda = \pm 1/2 \rightarrow (1, 1)$  dan  $(-1, -1)$ : **dua titik stasioner**, bedakan lewat kondisi cukup  
 4 variabel + 2 pembatas  $\rightarrow 4+2 = 6$  persamaan, 6 variabel.

## 3 · Hessian bordered

$$H^b = \begin{bmatrix} 0 & (m \times m) & P & (m \times n) \\ P^T & (n \times m) & Q & (n \times n) \end{bmatrix}$$

$P = \partial g_i / \partial x_j$   
 $Q = \partial^2 L / \partial x \partial x$

- Ukuran  $(m+n) \times (m+n)$ ; evaluasi di  $(x^*, \lambda^*)$
- $P: \partial g_i / \partial x_j$  ( $m \times n$ );  $Q$  = Hessian  $L$  terhadap  $x$  ( $n \times n$ )
- $n=4$ ,  $m=2 \rightarrow 6 \times 6$ ,  $P 2 \times 4$ ,  $Q 4 \times 4$

## 4 · Kondisi cukup: Hessian bordered

Periksa determinan  $n-m$  minor prinsipal (LPM) terakhir  $H^b$ , mulai dari **orde**  $2m+1$  sampai  $m+n$ .

|          | tanda det                          |
|----------|------------------------------------|
| maksimum | berganti-ganti, mulai $(-1)^{m+1}$ |
| minimum  | semua $(-1)^m$                     |

| n, m | orde | maks | min  |
|------|------|------|------|
| 2, 1 | 3    | +    | -    |
| 3, 1 | 3, 4 | +, - | -, - |
| 4, 2 | 5, 6 | -, + | +, + |

min slide ( $n=2, m=1$ ):  $\det \begin{bmatrix} 0 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} = -4$   
 $< 0 \rightarrow$  **min**

maks  $x_1 x_2$ :  $\det \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} = 2 > 0 \rightarrow$  **maks**

maks slide ( $n=3$ ):  $-\sum x^2 + 2\sum x$ ,  $\sum x = 1 \rightarrow x^* = (1/3, 1/3, 1/3)$ ,  $\lambda = 4/3$ ; orde 3 = 4, orde 4 = -12 (+, -)  $\rightarrow$  **maks**  
 Maks  $-\sum x^2$  dg  $x_1 + 2x_2 + 3x_3 = 14$ :  $x^* = (1, 2, 3)$ ,  $\lambda = -2$ , orde 3 = 10, orde 4 = -56  $\rightarrow$  maks.

## 5 · Kondisi cukup: metode $\Delta$

$$\Delta = [[0, P], [P^T, Q - \mu I]]$$

$\det(\Delta) = 0$   
 $I$  identitas  $n \times n$ ; hasil: polinomial dalam  $\mu$

| akar $\mu$    | $x^*$ adalah |
|---------------|--------------|
| semua negatif | maksimum     |
| semua positif | minimum      |

Contoh 1:  $\det(\Delta) = 2\mu - 4 = 0 \rightarrow \mu = 2 \rightarrow$  **min**

Contoh 2 (maks):  $\det(\Delta) = -3\mu^2 - 12\mu - 12 = 0 \rightarrow \mu = -2$  (dua kali)  $\rightarrow$  **maks**

Hasil slide Contoh 3 (min,  $f = \sum x^2 + 2\sum x$ ):  $H^b$  orde 3, 4 = -4, -12;  $\mu = 2$  (ganda)  $\rightarrow$  min. Periksa lagi nilai  $x^*$  dan  $\lambda$  di slide itu.

## 6 · Pembatas pertidaksamaan: KKT

Model: opt  $f(x)$  dengan  $g_i(x) \leq 0$ ,  $i=1..m$ .  $L = f - \sum \lambda_i g_i$ .

**Bentuk standar:**  $2x_1 - x_2 \geq 4 \rightarrow -2x_1 + x_2 + 4 \leq 0$  (ruas ke kiri, kalikan -1  $\rightarrow$  tanda berbalik)

| Syarat KKT              | bentuk                                              |
|-------------------------|-----------------------------------------------------|
| tanda $\lambda$         | $\lambda_i \geq 0$ (maks); $\lambda_i \leq 0$ (min) |
| stasioner               | $\partial L / \partial x_j = 0$ , $j=1..n$          |
| complementary slackness | $\lambda_i g_i(x) = 0$                              |
| kelayakan               | $g_i(x) \leq 0$                                     |

Aktif:  $g_i = 0$ ; tak aktif ( $g_i < 0$ )  $\rightarrow \lambda_i = 0$ .  
 Jumlah syarat =  $m+n+m+m$ :  $n=3, m=4 \rightarrow$  **15**; slide  $n=2, m=3 \rightarrow$  11

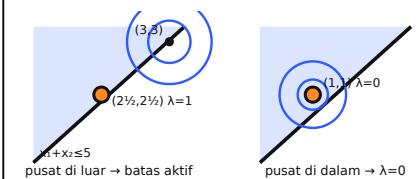
## 7 · Prosedur kasus KKT

- Bagi **2<sup>m</sup> kasus**: tiap  $\lambda_i = 0$  atau  $\neq 0$  ( $m=4 \rightarrow 16$ ;  $m=3 \rightarrow 8$ )
- $\lambda_i \neq 0 \rightarrow g_i = 0$  (aktif);  $\lambda_i = 0 \rightarrow$  abaikan  $g_i$
- Pecahkan  $x, \lambda$  dari stasioner + pembatas aktif
- Cek **layak** (semua  $g \leq 0$ ) dan **tanda  $\lambda$**
- Lolos keduanya = memenuhi kondisi perlu

| Soal                                               | Kasus            | x        | $\lambda$ | hasil                 |
|----------------------------------------------------|------------------|----------|-----------|-----------------------|
| maks $-(x_1-3)^2 - (x_2-3)^2$ , $x_1 + x_2 \leq 5$ | $\lambda = 0$    | (3, 3)   | 0         | tak layak ( $6 > 5$ ) |
|                                                    | $\lambda \neq 0$ | (2½, 2½) | 1         | lolos ( $\geq 0$ )    |
| maks $-(x_1-1)^2 - (x_2-1)^2$ , sama               | $\lambda \neq 0$ | (2½, 2½) | -1        | tanda salah           |
|                                                    | $\lambda = 0$    | (1, 1)   | 0         | lolos                 |
| min $x_1 + x_2$ , $x_1 \geq 1, x_2 \geq 1$         | 1, 1 aktif       | (1, 1)   | -1, -1    | lolos ( $\leq 0$ )    |
| maks $-(x_1-2)^2 - (x_2-2)^2$ , $x_1 + x_2 \leq 2$ | $\lambda \neq 0$ | (1, 1)   | 2         | lolos                 |
| min $(x_1-3)^2 + (x_2-1)^2$ , $x_1 + x_2 \leq 2$   | $\lambda \neq 0$ | (2, 0)   | -2        | lolos                 |

Slide 3 pembatas (8 kasus): maks  $-(x_1-1)^2 - (x_2-1)^2$ ,  $x_1 + x_2 \leq 5$ ,  $x \geq 0$ : hanya  $\lambda = (0, 0, 0)$  di (1,1) lolos; mis. (5,0)  $\lambda_1 = -8$ ,  $\lambda_3 = -10$  ditolak (maks butuh  $\lambda \geq 0$ ); (0,0)  $\lambda = (0, -2, -2)$  ditolak.

## 8 · Kondisi cukup KKT & ringkasan



| Kasus       | f       | himpunan pembatas |
|-------------|---------|-------------------|
| maksimisasi | cekung  | cembung           |
| minimisasi  | cembung | cembung           |

Fungsi linier = cembung dan cekung. Min  $x_1 + x_2$ ,  $x \geq 1$ : f linier + S cembung  $\rightarrow$  (1,1) **minimum global**.

|           | Persamaan                           | Pertidaksamaan               |
|-----------|-------------------------------------|------------------------------|
| pembatas  | $g=0$                               | $g \leq 0$                   |
| $\lambda$ | tanda bebas                         | $\geq 0$ maks / $\leq 0$ min |
| syarat    | $\partial L / \partial \lambda = 0$ | $\lambda g = 0, g \leq 0$    |
| cukup     | $H^b$ atau $\Delta$                 | konveksitas f & S            |